

LIME: Local Interpretable Model-agnostic Explanations

↳ finds local linear approximation

While the model, e.g. $f_{\theta}(x_1, x_2, x_3)$, may be very complex globally. It is easier to approximate it around the vicinity of a particular instance.

Step 1. Create neighborhood of a particular instance by introducing small perturbations

eg. $(0.7, 0.2, 0.3)$

→	$(0.72, 0.21, 0.34)$	}	x_1 x_2 $\vdots$ x_m
→	$(0.68, 0.22, 0.31)$		
→	$\vdots$		
→	$(0.71, 0.19, 0.26)$		

Step 2. make predictions for all the perturbations

$y_1, \dots, y_m$

Step 3. measure distance between the original instance x and the perturbed instances

$d_1 = d(x, x_1)$ → for continuous: Could be Euclidean

$d_2 = d(x, x_2)$ → for categorical: if the value is the same → 0
is different → 1

$\vdots$

$d_n = d(x, x_n)$

Step 4: Compute weights of each perturbed instances

intuition: the instances close to the original instance get more weights

$$w_i = \exp\left(-\frac{d_i^2}{\sigma^2}\right)$$

→ kernel width

a small σ : → instances close to the original instance get more attentions

a larger σ : → instances far from the original instance also get attentions

Step 5: fit weighted linear regression

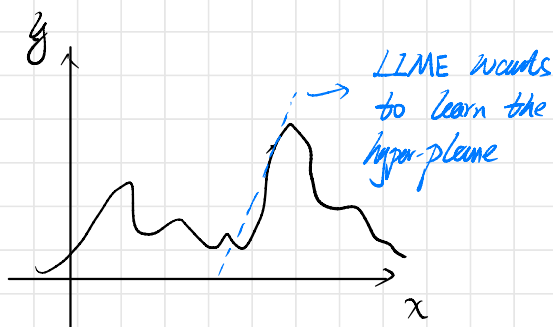
Input: $x_1, \dots, x_n$

Labels: $y_1, \dots, y_n$

loss: $\sum_{i=1}^n w_i \left(y_i - \sum_{j=1}^d (\beta + w_{ij}) \right)^2$

the weight w_j here tells us how the model output changes locally when the feature j changes

→ the loss for each y_i is weighted



For classification problem, how to interpret each feature for predicting →

$$\begin{aligned} p(A) &= 0.3 \\ p(B) &= 0.2 \\ p(C) &= 0.5 \end{aligned}$$

For explaining why the model predict $p(A) = 0.3$, do LIME

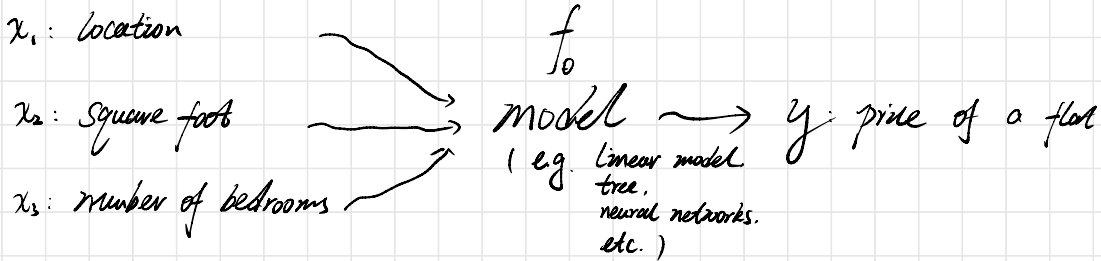
with $y = p(A)$ — i.e. take the probability $p(A)$ as the label

SHAP: SHapley Additive exPlanations

→ game theoretic approach to explain the output (prediction) of any machine learning model.

1. The "game" → to make one prediction
2. The "players" → model's features
3. The "payout" → final prediction - average prediction
4. The "question" → how to fairly distribute "payout" among different features?

Example:



Given x_1, x_2, x_3 , we obtain a prediction $\hat{y} = f_0(x_1, x_2, x_3)$.

How does each feature contribute to the prediction?

To find the contribution of x_i for the prediction $f_\theta(x_1, x_2, x_3)$

Consider: 1. Average prediction $E_{x_1, x_2, x_3}(f_\theta(x_1, x_2, x_3))$

all possible combinations: 2. prediction with only $\{x_1\}$, $E_{x_1, x_2, x_3}(f_\theta(x_1, x_2, x_3) | x_1 = x_1)$
 $\{x_2\}$,
 $\{x_3\}$.

3. prediction with $\{x_1, x_2\}$, $E_{x_1, x_2, x_3}(f_\theta(x_1, x_2, x_3) | x_1 = x_1, x_2 = x_2)$

$\{x_1, x_3\}$

$\{x_2, x_3\}$

$\{x_1, x_2, x_3\}$

Then:

Coalition Prediction without x_i

Prediction with x_i

marginal contribution of x_i

$\{\}$ $C(\{\}) E_{x_1, x_2, x_3}(f_\theta(x_1, x_2, x_3))$

$\{x_1\}$ $C(\{x_1\}) E_{x_1, x_2, x_3}(f_\theta(x_1, x_2, x_3) | x_1 = x_1)$

$C(\{x_1\}) - C(\{\})$

$\{x_2\}$ $C(\{x_2\}) E_{x_1, x_2, x_3}(f_\theta(x_1, x_2, x_3) | x_2 = x_2)$

$\{x_1, x_2\}$ $C(\{x_1, x_2\}) E_{x_1, x_2, x_3}(f_\theta(x_1, x_2, x_3) | x_1 = x_1, x_2 = x_2)$

$C(\{x_1, x_2\}) - C(\{x_1\})$

$\{x_3\}$ $\{x_3\}$

$\{x_1, x_3\}$

$\{x_2, x_3\}$

$\{x_2, x_3\}$

$\{x_1, x_2, x_3\}$

The SHAP value is then weighted average of each marginal contribution of x_i , where weight for each coalition is

n_i = number of features in coalition i

d = total features

weight

$$\frac{n_i! (d - n_i - 1)!}{d!}$$

for example: weight for coalition $\{x_2\}$: $\frac{1! (3 - 1 - 1)!}{3!} = \frac{1}{6}$

Above we compute SHAP value for one feature in one prediction.
Obviously we can compute SHAP value for all features
in many predictions.

→ Average SHAP values for one feature across many predictions ~ feature importance

practical consideration: evaluate 2nd marginal contributions
are impractical

→ need approximation method!

check `Shap.KernelExplainer`